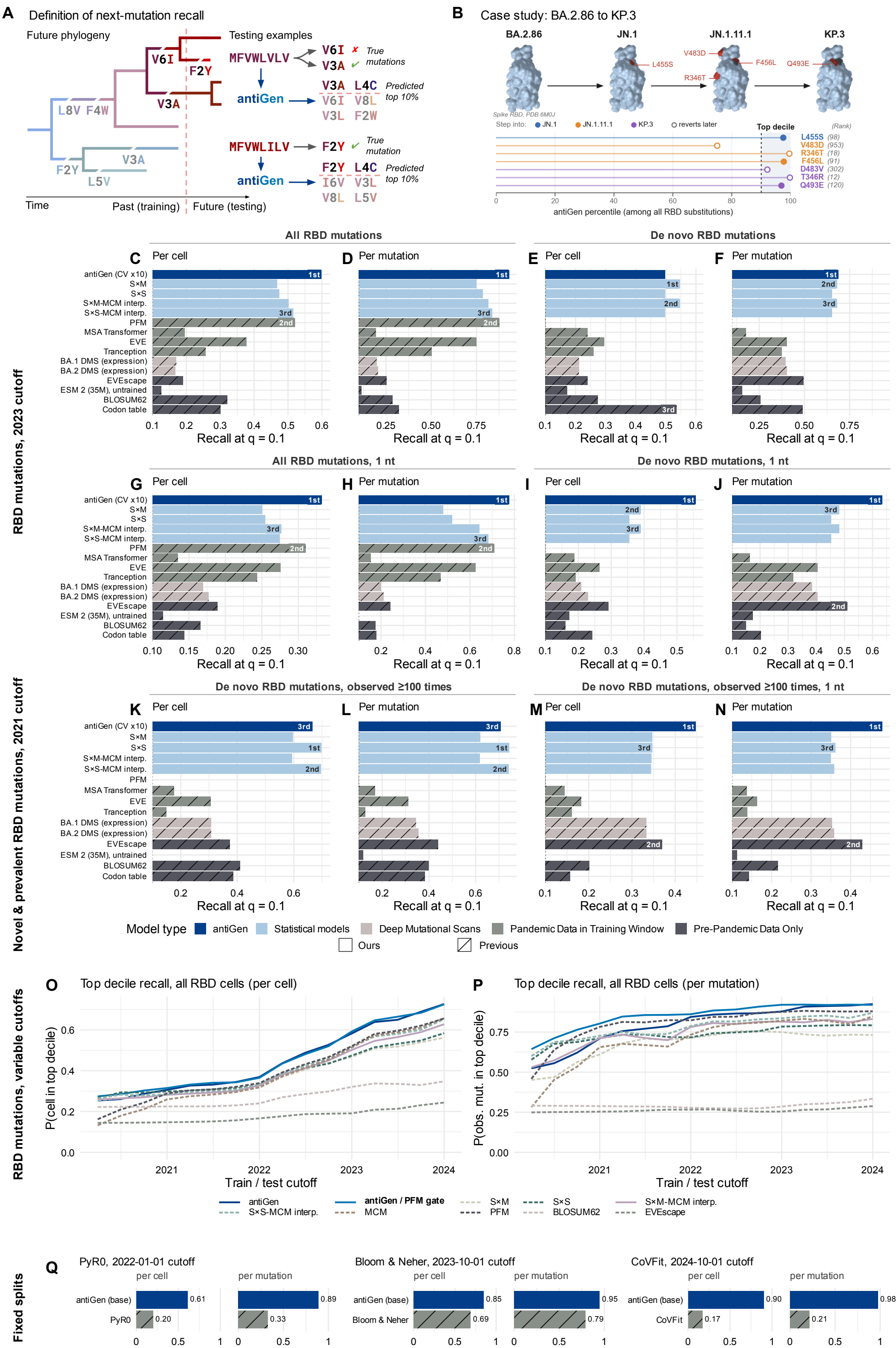

A
Definition of next-mutation recall
Future phylogeny
Testing examples
V6I
F2Y
V3A
L8V F4W
F2Y
V3A
L5V
MFVWLVLV
V6I ✗ True mutations
V3A ✓
antiGen
V3A L4C Predicted top 10%
V6I V8L
V3L F2W
MFVWLILV
F2Y ✓ True mutation
antiGen
F2Y L4C Predicted top 10%
I6V V3L
V8L L5V
Time
Past (training)
Future (testing)
B
Case study: BA.2.86 to KP.3
BA.2.86
JN.1
JN.1.11.1
KP.3
L455S
V483D
F456L
R346T
Q493E
Spike RBD, PDB 6M0J
Step into: JN.1 JN.1.11.1 KP.3 reverts later
Top decile
(Rank)
L455S (98)
V483D (953)
R346T (18)
F456L (91)
D483V (302)
T346R (12)
Q493E (120)
0 20 40 60 80 100
antiGen percentile (among all RBD substitutions)
All RBD mutations
De novo RBD mutations
C Per cell
D Per mutation
E Per cell
F Per mutation
antiGen (CV x10)
S×M
S×S
S×M-MCM interp.
S×S-MCM interp.
PFM
MSA Transformer
EVE
Tranception
BA.1 DMS (expression)
BA.2 DMS (expression)
EVEscape
ESM 2 (35M), untrained
BLOSUM62
Codon table
1st
2nd
3rd
0.1 0.2 0.3 0.4 0.5 0.6
0.25 0.50 0.75
Recall at q = 0.1
All RBD mutations, 1 nt
De novo RBD mutations, 1 nt
G Per cell
H Per mutation
I Per cell
J Per mutation
0.10 0.15 0.20 0.25 0.30
0.2 0.4 0.6 0.8
0.1 0.2 0.3 0.4 0.5
0.1 0.2 0.3 0.4 0.5 0.6
De novo RBD mutations, observed ≥100 times
De novo RBD mutations, observed ≥100 times, 1 nt
K Per cell
L Per mutation
M Per cell
N Per mutation
0.2 0.4 0.6
0.1 0.2 0.3 0.4
RBD mutations, 2023 cutoff
RBD mutations, 2021 cutoff & prevalent RBD mutations, 2021 cutoff
Novel
Model type
antiGen
Statistical models
Deep Mutational Scans
Pandemic Data in Training Window
Pre-Pandemic Data Only
Ours
Previous
O Top decile recall, all RBD cells (per cell)
P Top decile recall, all RBD cells (per mutation)
P(cell in top decile)
P(obs. mut. in top decile)
0.0 0.2 0.4 0.6
0.00 0.25 0.50 0.75
2021 2022 2023 2024
Train / test cutoff
antiGen
antiGen / PFM gate
S×M
S×S
S×M-MCM interp.
S×S-MCM interp.
MCM
PFM
BLOSUM62
EVEscape
Fixed splits
Q
PyR0, 2022-01-01 cutoff
Bloom & Neher, 2023-10-01 cutoff
CoVFit, 2024-10-01 cutoff
per cell
per mutation
antiGen (base)
PyR0
Bloom & Neher
CoVFit
0.61
0.89
0.20
0.33
0.85
0.95
0.69
0.79
0.90
0.98
0.17
0.21
0 0.5 1
Recall at q = 0.1